

Dongmin (Eugene) Bang (he/him)

Pharm.D, Ph.D in Bioinformatics

PERSONAL INFORMATION

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Google Scholar: <https://scholar.google.com/citations?user=9R0mnIAAAAAJ>

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HIGHLIGHTS

- **PharmD and PhD in Bioinformatics** at Seoul National University, awarded with **Youlchon AI Young Researcher** distinction, an annual recognition granted to ~10 students for outstanding interdisciplinary research in AI.
- Positioned at the intersection of systems pharmacology and machine learning, my research focuses on **knowledge-guided and data-driven modeling of biological interactions** to enable AI-augmented precision medicine and therapeutic design.
- Contributed to 16 peer-reviewed publications (9 first/co-first author) during doctoral study, including **first-author papers in *Nature Communications*, *ICML*, and *ISMB***, on integrative and multi-modal modeling of knowledge graphs, drug perturbation responses, and representation learning across biomedical scales to simulate systemic therapeutic responses.
- Completed a **competitive, university-funded research grant** (Advanced Medical Researcher Training Program–Seoul National University), leading an independent project on foundational modeling for virtual drug-induced cell state transition and multi-scale drug response prediction, resulting in a **first-authored ISMB 2024 publication**.
- Mentored interns and junior researchers, contributing to **four co-first-author publications in *Journal of Cheminformatics* and *ISMB 2025***, and multiple collaborative works across multi-modal representation learning, molecular modeling, and network biology.
- Led an **AI-driven drug discovery team** in industry, collaborating with hospitals, biotech startups, and medicinal chemistry groups to integrate omics-informed modeling, perturbation response simulation, and medicinal chemistry insights for *in silico* target prioritization, response prediction and candidate optimization for small molecules, antibodies and PROTACs.

EDUCATION

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| 03/2021–Present | Ph.D in Bioinformatics
Seoul National University, Seoul, Republic of Korea.
Advisor: Dr. Sun Kim
Dissertation: <i>Modeling Multi-Scale Biomedical Interactions via Transfer Learning and Knowledge Integration for Personalized Medicine</i> |
| 03/2015–02/2021 | Pharm.D, College of Pharmacy
Chung-Ang University, Seoul, Republic of Korea. |
| 03/2012–02/2015 | B.S.E student in Architectural Engineering
Hanyang University, Seoul, Republic of Korea. |

PROFESSIONAL EXPERIENCES

Research Experience

09/2022–Current **AIGENDRUG Co., Ltd.**, Seoul, Republic of Korea
Position: Senior Research Scientist, AI-Drug Discovery Team
Role: Led an AI-driven drug discovery team, developing AI-integrated pipelines for DrugVLAB™, an in-house multi-omics platform for drug design, target identification and drug response prediction. Coordinated cross-functional collaborations with academia, hospitals, biotech, and pharmaceutical partners on projects spanning AI-based drug design, Lab-in-the-loop AI, antibody optimization, and omics-driven drug response prediction. Invented and patented two novel AI frameworks for drug discovery (ROK patents, granted).

SELECTED PUBLICATIONS (Full list available in the Appendix)

(†: Equal contributions)

1. **Bang, D.**, Lim, S., Lee, S., & Kim, S. (2023). Biomedical knowledge graph learning for drug repurposing by extending guilt-by-association to multiple layers. *Nature Communications*.
- **Contributions:** Design and implementation of semantic teleport-based random walk for learning biomedical knowledge graph and case studies.
2. **Bang, D.**, Sung, I., Piao, Y., Lee, S., & Kim, S. (2025). Predicting Drug-likeness via Biomedical Knowledge Alignment and EM-like One-Class Boundary Optimization. *ICML 2025*
- **Contributions:** design and implementation of multi-modal alignment and EM-like optimization of drug-likeness boundary and case studies.
3. **Bang, D.**[†], Koo, B.[†], & Kim, S. (2024). Transfer learning of condition-specific perturbation in gene interactions improves drug response prediction. *Bioinformatics*–Transferred from *ISMB 2024*
- **Contributions:** Design and implementation of transfer learning pipeline between perturbational scales (gene to cell) and the condition-specific gene-gene attention.
4. Kim, Y.[†], **Bang, D.**[†], Koo, B., Yi, J., Cho, C., Choi, J., & Kim, S. (2025). MixingDTA: Improved Drug-Target Affinity Prediction by Extending Mixup with Guilt-By-Association. *Bioinformatics*–Transferred from *ISMB/ECCB 2025*
- **Contributions:** Design of multi-modal LLM integration, Guilt-by-association based Mixup and case studies. Leading lab juniors for experimental setup.
5. Lee, J.[†], **Bang, D.**[†], & Kim, S. (2024). Residue-Level Multiview Deep Learning for ATP Binding Site Prediction and Applications in Kinase Inhibitors. *Journal of Chemical Information and Modeling*, 65(1), 50-61.
- **Contributions:** design of protein LLM with 3D geometric GNN integration, and kinase case studies, leading intern for experimental setup.

AWARDS, HONORS & GRANTS

Aug. 2025 **Youlchon AI Young Researcher Award:** An annual recognition granted to ~10 students each year for outstanding research in AI applications (AI+X).

2023–2024	Research Grant: Advanced Medical Researcher Training Support Program (<i>Seoul National University</i>), Project Title: <i>Predicting Patient Drug Response with AI Using Transfer Learning and Omics Data</i>
Nov. 2024	Best Paper Award , Annual Conference of Korean Society for Bioinformatics (BIOINFO) 2024, Gyeongju, Republic of Korea
Nov. 2023	Best Paper Award , Annual Conference of Korean Society for Bioinformatics (BIOINFO) 2023, Yeosu, Republic of Korea
Aug. 2022	Honorable Mention at The AI Korea 2022 poster presentation, KIISE Artificial Intelligence Society, Republic of Korea
Jul. 2022	Honorable Mention at MOGAM Artificial Intelligence Symposium poster presentation, MOGAM Institute for Biomedical Research, Republic of Korea
Jun. 2022	Grand Prize in Artificial Intelligence , Korea Computer Congress (KCC) 2022, Jeju, Republic of Korea (Top 1/391 papers)

PRESENTATIONS

Jul. 2024	Oral Presentation , ADME-Drug-Likeness: Enriching Molecular Foundation Models via Pharmacokinetics-Guided Multi-Task Learning for Drug-likeness Prediction. <i>ISMB/ECCB 2025, Liverpool</i>
Jul. 2024	Oral Presentation , MixingDTA: Improved Drug-Target Affinity Prediction by Extending Mixup with Guilt-By-Association <i>ISMB/ECCB 2025, Liverpool</i>
Jul. 2025	Poster Presentation , Predicting Drug-likeness via Biomedical Knowledge Alignment and EM-like One-Class Boundary Optimization. <i>International Conference on Machine Learning (ICML) 2025, Vancouver</i>
Jul. 2024	Oral Presentation , Transfer learning of condition-specific perturbation in gene interactions improves drug response prediction. <i>International Conference on Intelligent Systems for Molecular Biology (ISMB) 2024, Montréal, Canada</i>
Jul. 2024	Invited Talk , Transfer learning of condition-specific perturbation in gene interactions improves drug response prediction. <i>Drug Discovery AI Workshop, Korea Computer Congress (KCC) 2024, Jeju, Republic of Korea</i>
Nov. 2023	Guest Lecturer , Introduction to Cheminformatics and databases & Network representation learning with word2vec. <i>AI-BIO Researcher Training Course, Artificial Intelligence Institute (AIIS), Seoul National University</i>
Nov. 2023	Oral Presentation , Biomedical knowledge graph learning for drug repurposing by extending guilt-by-association to multiple layers. <i>Annual Conference of Korean Society for Bioinformatics (BIOINFO) 2023, Yeosu, Republic of Korea</i>
Jul. 2023	Invited Talk , Biomedical knowledge graph learning for drug repurposing by extending guilt-by-association to multiple layers. <i>Biological Research Information Center (BRIC), POSTECH</i>
Nov. 2022	Guest Lecturer , Disease networks and drug repurposing. <i>AI-BIO Researcher Training Course, Artificial Intelligence Institute (AIIS), Seoul National University</i>

TEACHING & MENTORSHIP

Teaching (Computer Science & Engineering Dept., Seoul National University)

Spring 2025	Teaching Assistant , Machine Learning on Bioinformatics (English Lecture) - Bioinformatics, Data & Machine Learning Algorithms for Cancer Classification - Multi-Layer Perceptron and Deep Learning with PyTorch and CoLab
Fall 2024	Teaching Assistant , Algorithms - Efficient string matching for whole-genome k-mer counting and sorting
Spring 2024	Teaching Assistant , Machine Learning on Bioinformatics (English Lecture) - Bioinformatics, Data & Machine Learning Algorithms for Cancer Classification - Multi-Layer Perceptron and Deep Learning with PyTorch and CoLab
Fall 2023	Teaching Assistant , Computer Convergence & Applications (English Lecture) - Introduction to AI drug discovery: Navigating the chemical space with AI technologies
Spring 2023	Teaching Assistant , Machine Learning on Bioinformatics (English Lecture) - Practical implementation of HMM for exon identification with Python - Molecule representation learning & property prediction with Deep Learning
Spring 2022	Teaching Assistant , Machine Learning on Bioinformatics (English Lecture) - Practical implementation of HMM for exon identification with Python

Mentorship

Spring 2025	Team Mentor: Development of RAG-Based AI Tool Selection and Interaction Platform for Drug Discovery, <i>Creative & Integrative Design Internship Course</i>
Fall 2024	Team Mentor: Real-time Ensemble Framework for Compound Activity Prediction from User Data Input, <i>Creative & Integrative Design Internship Course</i>
Spring 2024	Team Mentor: Integrating ADME data through multi-task learning for defining drug-likeness 2, <i>Creative & Integrative Design Internship Course</i> – Resulted in publication at <i>ISMB/ECCB 2025</i> .
Winter 2023	Mentor: Biological Prior Knowledge-integrated Pocket-based Drug Design, <i>Research Internship Program for Interdisciplinary Major in Artificial Intelligence</i>
Fall 2023	Team Mentor: Integrating ADME data through multi-task learning for defining drug-likeness 1, <i>Creative & Integrative Design Internship Course</i>
Spring 2023	Team Mentor: ATP-binding site prediction using protein Language Models and 3D structure, <i>Creative & Integrative Design Internship Course</i> – Resulted in publication at <i>Journal of Chemical Information and Modeling</i> .
Winter 2022	Mentor: Contrastive Learning Strategies for Molecular Property Prediction, <i>Research Internship Program for Interdisciplinary Major in Artificial Intelligence</i>
Fall 2022	Team Mentor: ATP-binding site prediction using 3D structure, <i>Creative & Integrative Design Internship Course</i>

COMMUNITY SERVICE

2022-2025	Reviewer Involvement
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(Conferences) *AAAI, ICML, ICLR, KDD*.

(Journal) *Nature Communications, Nature Machine Intelligence, Nature Biomedical Engineering, Briefings in Bioinformatics, Bioinformatics, Scientific Reports*.

TECHNICAL EXPERTISE

Machine Learning: PyTorch, PyTorch Geometric, HuggingFace Transformers, basic TensorFlow; Python-implemented machine learning algorithms and AutoML.

Bioinformatics & cheminformatics: RDKit, Biopython, DeepChem, TorchDrug, scanpy, basic R/Bioconductor

Data analysis & visualization: pandas, NumPy, anndata, seaborn, matplotlib

Last updated: September 2, 2025

APPENDIX

FULL PUBLICATIONS LIST (†: EQUAL CONTRIBUTIONS)

Statistics: 11 Journal Publications (5 first/co-firsts), 5 Conference Proceedings (4 first/co-firsts), 3 Under Review (1 first) since 2022, with ~190 citations (provided by Google Scholar).

Journal Publications

1. Kong, D.[†], Ha, Y.[†], Yoo, H.[†], **Bang, D.**, & Kim, S. (2025). Survey on AI-Drug Discovery with Knowledge Graphs: Data, Algorithm and Application. *Journal of Computing Science and Engineering* (Accepted)
2. Ha, S.[†], **Bang, D.**[†], & Kim, S. (2025). FATE-Tox: Fragment Attention Transformer for E(3)-Equivariant Multi-Organ Toxicity Prediction. *Journal of Cheminformatics*, 17, 74.
3. Sung, I., Lee, S., **Bang, D.**, Yi, J., Lee, S., & Kim, S. (2025). MDTR: a knowledge-guided interpretable representation for quantifying liver toxicity at transcriptomic level. *Frontiers in Pharmacology*, 15, 1398370.
4. Lee, J.[†], **Bang, D.**[†], & Kim, S. (2024). Residue-Level Multiview Deep Learning for ATP Binding Site Prediction and Applications in Kinase Inhibitors. *Journal of Chemical Information and Modeling*, 65(1), 50-61.
5. Cho, C.[†], Lee, S.[†], **Bang, D.**, Piao, Y., & Kim, S. (2024). ChemAP: predicting drug approval with chemical structures before clinical trial phase by leveraging multi-modal embedding space and knowledge distillation. *Scientific Reports*, 14(1), 23010.
6. Pak, M., **Bang, D.**, Sung, I., Kim, S., & Lee, S. (2024). DGDRP: drug-specific gene selection for drug response prediction via re-ranking through propagating and learning biological network. *Frontiers in Genetics*, 15, 1441558.
7. Gu, J.[†], **Bang, D.**[†], Yi, J.[†], Lee, S., Kim, D. K., & Kim, S. (2023). A model-agnostic framework to enhance knowledge graph-based drug combination prediction with drug-drug interaction data and supervised contrastive learning. *Briefings in Bioinformatics*, 24(5), bbad285.
8. **Bang, D.**, Lim, S., Lee, S., & Kim, S. (2023). Biomedical knowledge graph learning for drug repurposing by extending guilt-by-association to multiple layers. *Nature Communications*, 14(1), 3570.
9. Shin, J.[†], Piao, Y.[†], **Bang, D.**, Kim, S., & Jo, K. (2022). DRPreter: interpretable anticancer drug response prediction using knowledge-guided graph neural networks and transformer. *International Journal of Molecular Sciences*, 23(22), 13919.
10. **Bang, D.**[†], Gu, J.[†], Park, J., Jeong, D., Koo, B., Yi, J., ... & Lee, S. (2022). A survey on computational methods for investigation on ncRNA-disease association through the mode of action perspective. *International Journal of Molecular Sciences*, 23(19), 11498.
11. Lim, S.[†], Lee, S.[†], Piao, Y., Choi, M., **Bang, D.**, Gu, J., & Kim, S. (2022). On modeling and utilizing chemical compound information with deep learning technologies: A task-oriented approach. *Computational and structural biotechnology journal*, 20, 4288-4304.

Conference Proceedings

1. **Bang, D.**, Sung, I., Piao, Y., Lee, S., & Kim, S. (2025). Predicting Drug-likeness via Biomedical Knowledge Alignment and EM-like One-Class Boundary Optimization. *ICML 2025* (Accepted for poster presentation)
2. **Bang, D.**[†], Kim, J.[†], Song, H., & Kim, S. (2025). ADME-Drug-Likeness: Enriching Molecular Foundation Models via Pharmacokinetics-Guided Multi-Task Learning for Drug-likeness Prediction. *ISMB/ECCB 2025* (Accepted)

3. Kim, Y.[†], **Bang, D.**[†], Koo, B., Yi, J., Cho, C., Choi, J., & Kim, S. (2025). MixingDTA: Improved Drug-Target Affinity Prediction by Extending Mixup with Guilt-By-Association. *ISMB/ECCB 2025* (Accepted)
4. **Bang, D.**[†], Koo, B.[†], & Kim, S. (2024, July). Transfer learning of condition-specific perturbation in gene interactions improves drug response prediction. *ISMB2024* – Transferred to *Bioinformatics*, 40(Supplement_1), i130-i139.
5. Lee, D., Lee, D., **Bang, D.**, & Kim, S. (2024, March). DISCO: Diffusion schrödinger bridge for molecular conformer optimization. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 38, No. 12, pp. 13365-13373).

Preprints and Under Review

1. **Bang, D.**[†], Koo, B.[†], & Kim, S. Transfer Learning of Condition-Specific Perturbation in Gene Interactions: Towards Multi-modal Foundational Modeling of Drug Response. In *ICML 2025 Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences*
2. Sung, I., **Bang, D.**, Kim, S., & Lee, S. Transferring Preclinical Drug Response to Patient via Tumor Heterogeneity-Aware Alignment and Perturbation Modeling. In *ICLR 2025 Workshop on Machine Learning for Genomics Explorations*
3. **Bang, D.**, Sung, I., & Kim, S. Predicting Therapeutic Outcome via Aligning Patient-Specific Knowledge Graphs and Translating Gene-Level Perturbation Patterns. (Under review)
4. Song, H., **Bang, D.**, Koo, B., Kim, S., & Lee, S. LLM-Integrated Representative Path Selection for Context-Aware Drug Repurposing on Biomedical Knowledge Graphs (Under Review)
5. **Bang, D.**, Yoon, I., Sung, I., & Kim, S. Predicting Single-cell Response to Chemical Through Modeling Gene Interactions with Multi-modal Attention (Under Review)
6. Kim, J., **Bang, D.**, & Kim, S. Active Learning for Hit Compound Identification via Deep One-class Boundaries on Multiple clusters. (Under Review)